# EPMA TAPP review workbook

This workbook asks you to review the EPMA TAPP: the list of metadata items that should be reported when an EPMA procedure is registered, and when a measurement session run under that procedure is documented. It has four tabs: **Instructions** (this tab), **Reviewer information**, **EPMA TAPP**, and **Registration Metadata**. Please review the 78 metadata items on the **EPMA TAPP** tab. The Registration Metadata tab lists 14 items that are set by registration requirements; they need no review, though you may comment on them in "Other comments".

Every item is described for two records. The procedure record is the standing recipe, registered once and cited by DOI. The session record describes one measurement session run under that procedure; a session may cover several samples. **Note: A TAPP is not a procedure or session record, but guidelines for what metadata is required when reporting a procedure/session.**

## What to do

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|---|---|
| **Step 1** | **Fill in the "Reviewer information" tab.** Answers go in the "Your answer" column; the question column is locked. |
| **Step 2** | **On the "EPMA TAPP" tab, put all your feedback in the six shaded comment columns**: "Needed?", "Structure OK?" (twice), "Tier OK?" (twice) and "Other comments". Each one judges the information column on its left; "Other comments" takes everything else. Leave a cell blank if you agree or have no view. All other columns are locked. |
| **Step 3** | **Follow the "Review criteria"** at the bottom of this tab. Provide review that clearly presents your **suggested change and reasons**. |
| **Step 4** | When you have finished, **fill in the review summary on the "Review Information" tab. Save the workbook as EPMA_TAPP_Review_<your initials>.xlsx** (e.g., EPMA_TAPP_Review_RD.xlsx) and **email it to info@astromat.org**. The deadline for this round of review is **December 1, 2026**. |

## The EPMA TAPP tab: column explanations

The tab has 12 information columns and 6 comment columns. The information columns are: **"Index"**, **"Metadata item"**, **"Description"**, **"Example (NOT recommendations)"**, **"Structure — procedure record"**, **"Procedure-Level Tier"**, **"Structure — session record"**, **"Session-Level Tier"**, **"EDS Point Analysis"**, **"EDS Mapping"**, **"WDS Point Analysis"**, **"WDS Mapping"**. The comment columns are: **"Needed?"**, **"Structure OK?"** (twice), **"Tier OK?"** (twice) and **"Other comments"**. Each comment column sits immediately to the right of the information column it judges and has the same colour.

| Information Column | Meaning |
|---|---|
| **Index** | The item's number in the TAPP: group number, then position in the group. |
| **Metadata item** | The name of the metadata item. |
| **Description** | What the item records. |
| **Example (NOT recommendations)** | An illustration of the item filled in, from a fictional worked example (an invented EPMA session written up as a methods section). It shows the shape of an answer. It is not recommended content, and the laboratory, samples and results are invented. Samples are CC-1 and CC-2; analysis points are named by mineral and number (ol3 = the third olivine point, ap2 = the second apatite point); maps are Map 1 and Map 2. |
| **Structure — procedure record** | How the procedure states the item: one entry, one entry for each entry of a list, or a list of its own. The session record follows it, as shown in the session structure column. |
| **Procedure-Level Tier** | Basic = mandatory for procedure registration. Advanced = recommended but optional. N/A = not part of the procedure: it can only be known once a session has run. |
| **Structure — session record** | How one measurement session reports the item. |
| **Session-Level Tier** | Read-Only = must follow the entry given in the procedure. Editable = may vary within the range the procedure allows; the session records what was used. Basic = the entry comes from the session itself and must be reported. Advanced = the entry comes from the session itself, and reporting it is recommended. |
| **EDS Point Analysis, EDS Mapping, WDS Point Analysis, WDS Mapping** | The analytical modes of EPMA. A tick means the item applies to that mode. |

## The EPMA TAPP tab: "Tiers"

| Tier | Meaning |
|---|---|
| **Basic** | Required |
| **Advanced** | Recommended |
| **Read-Only** | Fixed by the procedure. Session-level only. |
| **Editable** | Adjustable within the bounds the procedure sets. Session-level only. |
| **N/A** | Not applicable at that level. Procedure-level only. |

## The EPMA TAPP tab: "Structure"

Each **"Structure"** cell is an instruction, written in one of the following patterns. An **entry** is one complete answer to the item, in whatever form its "Description" asks for: a number with its unit, a name, a choice, or a short statement with several parts (for example a reference material with its source and a citation). The parts of one entry are given together, not as separate entries. A **list** is made of entries too, one per member, and other items refer to its members as entries.

| Structure Syntax | Meaning and Example |
|---|---|
| **List the …** | **List all appropriate entries.** Other metadata items may enumerate by the list. Example: Target Material (2.1) has the "structure - procedure record" of "List the material types the procedure is designed to analyse." Beam Current (4.4) then enumerates by Target Material (2.1). |
| **Give one entry for each X.** | **One entry for every entry in the list X.** "For the procedure" or "for the session" means one response per record. "One or more entries" means several may be given per record. Example: Beam Current (4.4) has the "structure - procedure record" of "Give one entry for each 'Target Material' (2.1)." |
| **For each A, give one entry for each B.** | **Two independent lists crossed: one entry for every A-B pairing, with A as the outer list.** Example: "Secondary Reference Material" (6.2) and "Reported Variable" (4.14) are independent lists. Analytical Precision (6.8) has the "structure - procedure record" of "For each 'Secondary Reference Material' (6.2), give one entry for each 'Reported Variable' (4.14)." |
| **For each A, give one entry for each of its B entries.** | **One entry for each B nested inside and only inside A.** Example: Each "Sampling Unit Name" (2.5) must have a corresponding "Sample Name" (2.3). Mapping Beam Mode (4.8) has the "structure - session record" of "For each 'Sample Name' (2.3), give one entry for each of its 'Sampling Unit Name' (2.5) entries." |
| **Give one entry; each "Sample Name" entry in a session inherits it.** | **Give one entry in the procedure record, and every sample (or point) in a session takes it.** Example: Sample Preparation Method (2.2) has the "structure - procedure record" of "Give one entry; each "Sample Name" (2.3) entry in a session inherits it." |

## Review criteria per column

| Comment Column | Review Criteria |
|---|---|
| **Needed?** | Is this item relevant to EPMA as you practise it? Is it one thing, and distinct from other items? Is it correctly named? |
| **Structure OK? (procedure)** | Does the procedure need one entry, one entry for each entry of a list (for example, per "Target Material"), or a list of its own? |
| **Tier OK? (procedure)** | Can it be known before any session? If not: "N/A". If so, is it essential for judging data quality or reproducing the work? Then: "Basic"; otherwise: "Advanced". |
| **Structure OK? (session)** | Does a session report it once, or once per sample, per analysis point, per element…? |
| **Tier OK? (session)** | Must the session copy the procedure's entry unchanged (Should be fixed by the procedure, i.e. "Read-Only")? May it vary within the procedure's bounds (Should be adjustable, i.e. "Editable")? Or is it known only from the session ("Basic"/"Advanced")? |
| **Other comments** | Anything else: the wording of Description or Example, missing items, wrong or missing modes, suggestions on the TAPP's structure. |

This workbook is developed by the Astromaterials Data System (astromat.org) team at the Lamont-Doherty Earth Observatory of Columbia University.